

DataKaleidos

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Chapter 1

Introduction

Cloud-based data platform that provides real-time, keyword-driven insights into market trends, consumer behavior, and competitor activities. The platform will automate collection of data from online sources and perform on-demand analysis. The web application will offer access to personalized reports, visualizations and an AI-driven chat-bot that helps users query specific data. [1].

1.1 Problem Statement

Businesses need to make data-driven decisions. This is an essential part of optimizing marketing and investment strategies. Many businesses face a critical lack of business intelligence regarding consumer sentiments. This leads to ill-informed decisions and inefficient usage of resources. Comprehensive market surveys must be carried out before new ventures, particularly in the Consumer Packaged Goods (CPG) industry.

This issue is particularly critical when it comes to small-scale businesses and start-ups that may not have the time or resources required to carry out these comprehensive surveys. In order to make wise investments and efficiently allocate funds, finding insights into the existing industries is crucial.

Although the convenience and popularity of the internet has made it easier to connect with consumers globally, and has provided an accessible platform for users to express their thoughts over a vast array of subjects, including consumer product goods, the relevant data is still scattered over multiple sites. A comprehensive solution to automate the gathering and cataloguing of this data in order to provide market insights would greatly improve the process.

Even with ready-made data sets and large-scale surveys that ask the right questions, performing custom analysis for user queries still poses a challenge. Navigating large data

sets manually requires expertise that may be difficult to obtain, especially over different existing solutions. A natural language system to query this data would be a novel solution. [2]

1.2 Scope

The scope of this project encompasses a cloud-based platform for business data analytics. The focus of the project will be the Consumer Packaged Goods (CPG) industry, in particular the Personal Care and Beauty industry. It will help businesses monitor market trends and understand consumer behavior, in real-time.

The process of collecting and cataloguing online data will be automated. Sources like social media, e-commerce sites, blogs, forums, and articles will be used. Regular updates to the data will ensure the platform stays effective and relevant in a rapidly changing market.

The collected data can then be processed and further analysed. This eliminates the need for manual survey and analysis, and provides quick access to real-time query-able insights. The platform will also generate reports, visualizations, and trend analysis.

An AI-powered chat-bot will allow users to ask questions and get instant responses, making data easy to access and understand. The platform's design will be simple and intuitive, so users of all technical backgrounds can navigate it easily.

To support large volumes of data, the platform will use secure, scalable cloud storage.

1.3 Modules

Modules of the proposed project are outlined below with features highlighted in numbered notation.

1.3.1 Module 1 - Web Application

Web application interface for intuitive access to the system.

1. Registration and authentication functionality to prevent unauthorized access to the system.
2. Search and Query (Chat-bot) Interface.
3. Data Visualisation Interface.
4. User Query Management and History.

1.3.2 Module 2 - Data Collection

Collection of data about user sentiments from various online sources.

1. Catalogue the collected data by brands and product types in the Personal Care and Beauty Sector.
2. Social Media API Integration to include social media posts and engagement in our dataset.
3. Web scraping from online user-written blogs, articles, and forums.
4. Web scraping customer reviews from major e-commerce platforms both local (e.g., Daraz.pk) and international (e.g., Amazon, Shopify).

1.3.3 Module 3 - Data Pre-processing

Preparing the collected raw data for analysis in the next stages.

1. Text Cleaning to normalize, remove noise and punctuation, and fix grammatical errors.
2. Tokenization to enable sentiment analysis.

3. Stemming and lemmatization.
4. Handling of irregular elements such as abbreviations, slang and emoticons.
5. Discarding duplicate, spam, or non-english language material.

1.3.4 Module 4 - Sentiment Analysis

Using Natural Language Processing to provide various types of sentiment analysis: fine-grained, aspect-based, intent-based and emotional detection.

1. Fine-tuning pre-trained LLMs for greater accuracy.
2. Sentiment Scoring on gathered and processed data.
3. Aspect-based analysis to evaluate specific features of products.
4. Topic and Keyword Extraction to identify important themes, keywords, and phrases related to products.
5. Intent-Based Analysis to determine the user's intent behind a review.
6. Real-time analysis of incoming data.

1.3.5 Module 5 - Data Visualization

Graphically represent the data for easier interpretation and to highlight patterns and trends.

1. Display average sentiment trends over time with line graphs.
2. Display proportions of positive and negative sentiments with pie charts.
3. Uncover keywords associated with your search elements with heat-maps.

1.3.6 Module 6 - Report Generation

Although the user may query the data using the chat-bot, the system will also generate a complete report after analysing all the data.

1. Key statistics and textual feedback highlighting important discoveries.
2. Report exportability as PDF downloads.

1.3.7 Module 7 - Chat-bot Development

This will allow the user to request specific data via natural language queries.

1. Dynamic processing of the user queries.
2. Natural language understanding to interpret user queries.
3. Intent Recognition to understand user's intent; to query sentiment, comparison, or trends related to data.
4. Follow-Up on queries for multi-turn conversations.

1.3.8 Module 8 - Cloud Integration

The data being collected and processed will be stored on the cloud for greater efficiency and scalability.

1. Setup cloud infrastructure to store raw and processed data.
2. Construct ETL (Extract, Transform, Load) pipeline to automate the process.
3. Deploy containers on the cloud to attain greater scalability.

1.4 User Classes and Characteristics

User class	Description
Business Owner	In the CPG industry, a Business Owner is a company's executive/product manager who wishes to use the platform to acquire real-time insights into consumer experiences and trends. They may frequently interact with the platform, especially during product launches, brand health monitoring, crisis management, competitive analysis, and strategic decision-making. Business Owners will use the chat-bot interface to quickly query consumer sentiment during meetings or on the go via desktop or laptop for high-level overviews and analytics.
Marketing Team	The Marketing Team comprises professionals dedicated to optimizing campaigns based on consumer sentiments within the CPG sector. They are expected to use the platform daily to monitor product effectiveness and assess the performance of marketing campaigns. The majority will access the platform via corporate desktop systems to track marketing performance, and adjust their strategies accordingly.
Investor	An investor is an organization or individual with capital who utilizes the platform to assess market trends, enabling them to make informed investment decisions within the CPG industry. They may periodically interact with the chatbot interface to analyze sentiments and consumers' intent through the desktop interface.
Data Analysts	A data analyst gathers, cleans, and interprets data sets for a specific purpose. They will use the insights generated by the system to supplement their own analysis.
Industry Researchers	Industry researcher use scientific methods to solve problems in the marketplace and develop innovative solutions as per the market demand. They will use the system's trend analysis features to conduct their market research.
Consumers	The customers who actually use the products in the CPG industry are the ones who will benefit most from a deeper understanding of the conversational volume surrounding their potential purchases. Instead of manually searching for reviews which may be tedious and un-insightful, a consumer will use prompt the system for suggestions before making a decision.

Table 1.1: User Classes

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-case Diagram

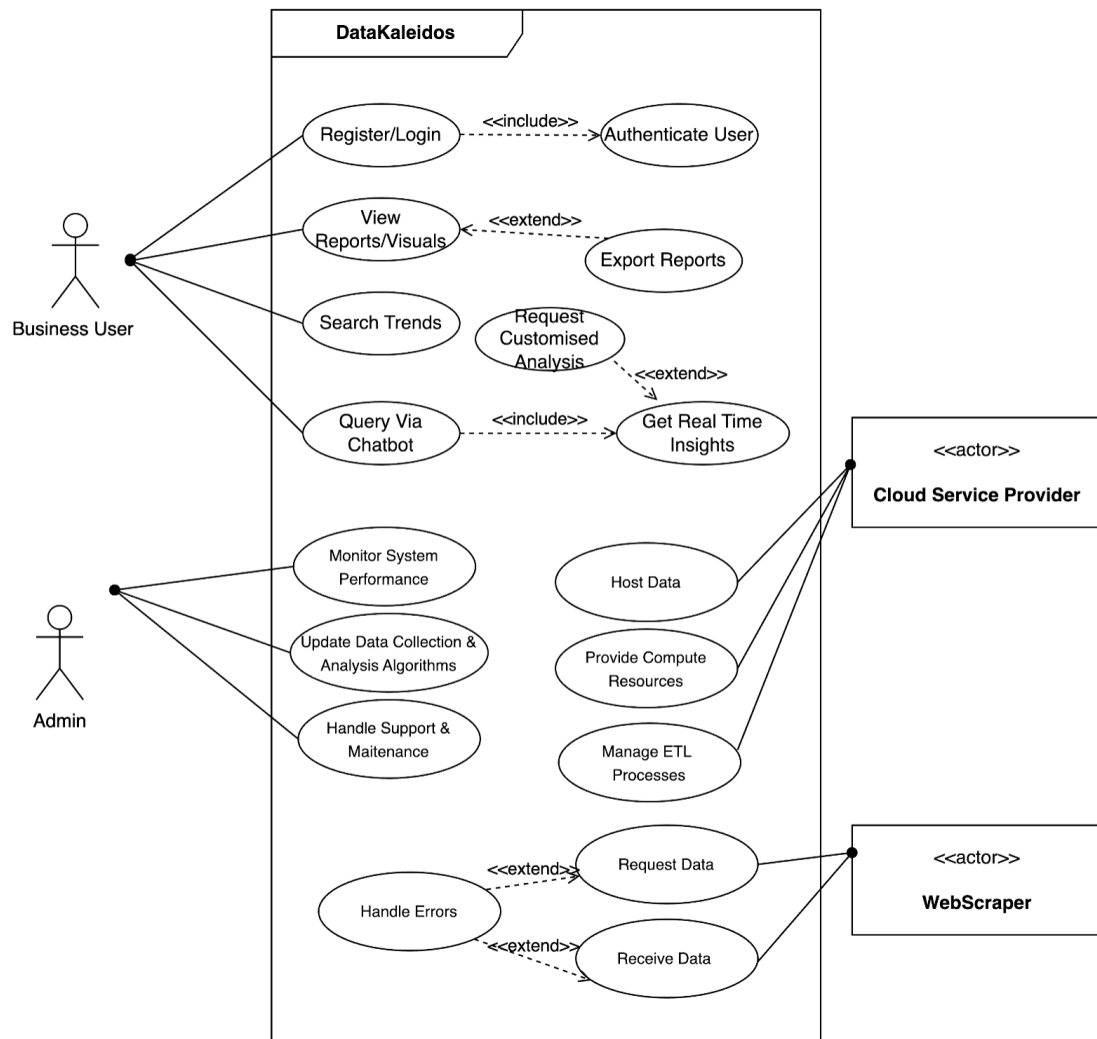


Figure 2.1: Use-case Diagram

2.2 Functional Requirements

Given are the functional requirements of the system in natural language. This section is organized by feature (module-wise) with functional requirements associated for each feature.

2.2.1 Module 1 - Web Application

1. The system shall allow users to register by providing an email, username, and password.
2. The system shall authenticate users based on their login credentials before granting access.
3. The system shall provide a search interface where users can query the data using keywords.
4. The system shall display visualizations of the queried data in the form of graphs and charts.
5. The system shall maintain a query history for each user, allowing them to view previous searches.

2.2.2 Module 2 - Data Collection

1. The system shall collect product reviews from online sources.
2. The system shall catalog collected data by brand and product type in the Personal Care and Beauty sector.
3. The system shall update the dataset regularly by running the collection process at least once every 24 hours.

2.2.3 Module 3 - Data Pre-processing

1. The system shall clean the raw data by removing punctuation, fixing grammatical errors, and normalizing the text.
2. The system shall tokenize the text data to prepare it for sentiment analysis.
3. The system shall apply stemming and lemmatization to reduce words to their root forms.

4. The system shall handle irregular data elements such as abbreviations, slang, and emoticons.
5. The system shall discard duplicate entries, spam, and non-English content before analysis.

2.2.4 Module 4 - Sentiment Analysis

1. The system shall perform sentiment analysis (score negative and positive sentiments) for each text entry.
2. The system shall identify specific product features mentioned in user comments.
3. The system shall extract topics and keywords related to products from user comments and reviews.

2.2.5 Module 5 - Data Visualization

1. The system shall generate line graphs showing average sentiment trends over a selected time period.
2. The system shall display pie charts representing the proportions of positive, negative, and neutral sentiments for a selected product.
3. The system shall provide heatmaps highlighting key topics and phrases.

2.2.6 Module 6 - Report Generation

1. The system shall generate reports summarizing key statistics, trends, and insights based on user queries.
2. The system shall allow users to export reports as PDF documents.

2.2.7 Module 7 - Chat-bot Development

1. The system shall allow users to interact with a chat-bot using natural language queries.
2. The system shall understand and process user queries to return relevant data and insights.
3. The system shall support multi-turn conversations by allowing users to follow up on previous queries without repeating information.

2.2.8 Module 8 - Cloud Integration

1. The system shall store all collected and processed data on a secure cloud platform.
2. The system shall implement an ETL (Extract, Transform, Load) process to automate data flow from collection to storage.
3. The system shall deploy the platform using cloud containers to ensure scalability and high availability.

2.3 Non-Functional Requirements

Non-functional requirements for the system are given below.

2.3.1 Reliability

RELI-1: To maintain reliability, the system will implement automated logging to detect anomalies.

RELI-2: In case of system failure, data processing should resume from the last successful step within 10 minutes after recovery.

RELI-3: The platform should handle errors gracefully, showing clear messages to users and logging issues without crashing.

RELI-4: The sentiment analysis module should consistently provide accurate results with at least 75-90% reliability, ensuring meaningful insights from the user reviews.

2.3.2 Usability

USE-1: The graphical user interface (GUI) shall exhibit an intuitive design and simple navigation, adopting a user-centered approach.

USE-2: Throughout the application, a consistent design and interaction pattern shall be maintained to enhance user familiarity

USE-3: 90% of users should be able to understand and interact with the dashboard within 10 minutes without needing training.

USE-4: Users should be able to configure and generate analytics reports within 3 clicks.

2.3.3 Performance

PER-1: The dashboard should load all interactive charts and analytics data within 5 seconds for datasets containing up to 1,000 entries.

PER-2: Users should be authenticated within 2 seconds of submitting their login credentials.

PER-3: PDF reports should be generated and downloadable within 5 seconds after the user requests them, for datasets of up to 1,000 entries.

PER-4: User chat history should load within 10 seconds when accessing any previously initiated conversation.

2.3.4 Security

SEC-1: : Robust user authentication and authorization mechanisms shall be implemented to prevent unauthorized access to user chat histories.

SEC-2: User data, including emails and passwords, must be securely protected using encryption.

2.3.5 Scalability

SCA-1: The platform should be able to process data sets containing up to 5,000 entries with consistent performance.

SCA-2: The architecture must support adding new modules (e.g., data sources) with minimal changes for future scalability.

SCA-3: The system should efficiently manage datasets up to 0.5 million CPG records, specifically Personal Care and Beauty, for smooth data fetch and visualization.

2.4 Domain Model

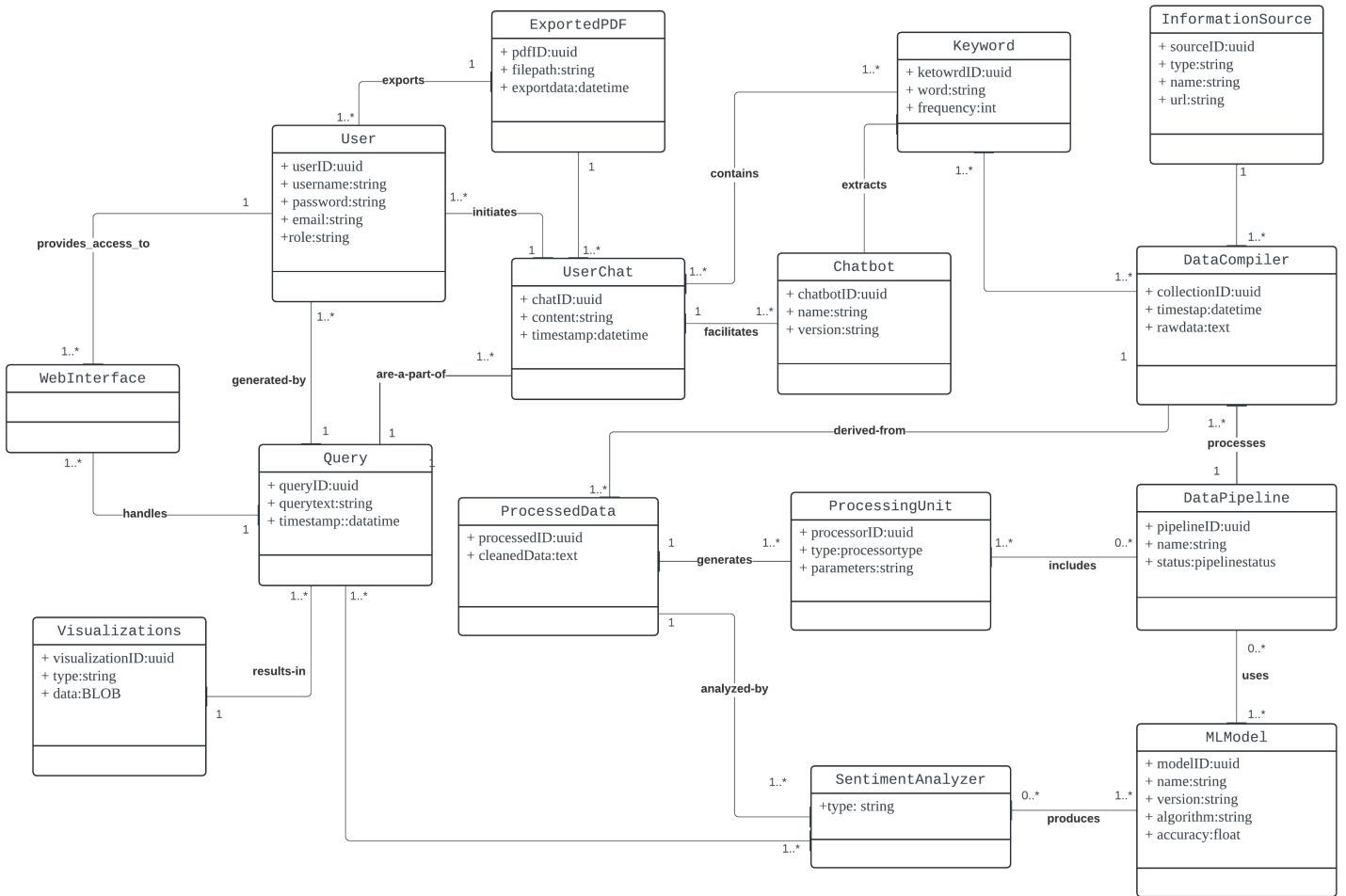


Figure 2.2: Domain Model Diagram

Chapter 3

System Overview

The system provides insights into market trends and customer sentiment. It is designed using a microservices architecture, which allows for modular, scalable, and manageable development. Each system component is implemented as an individual service within a cloud environment.

3.1 Architectural Design

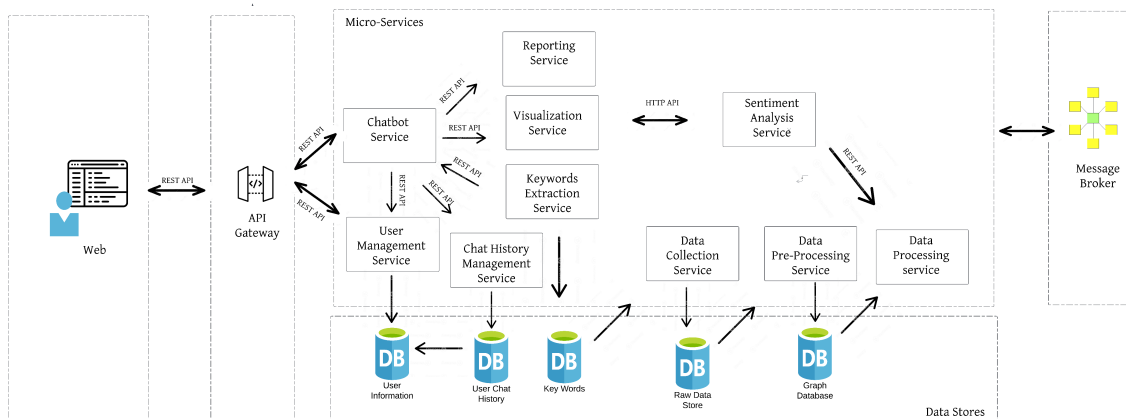


Figure 3.1: Micro-service Architecture Diagram

3.2 Design Models

Following are the design models as applicable to our system along with detailed descriptions.

Design Models for Procedural Approach

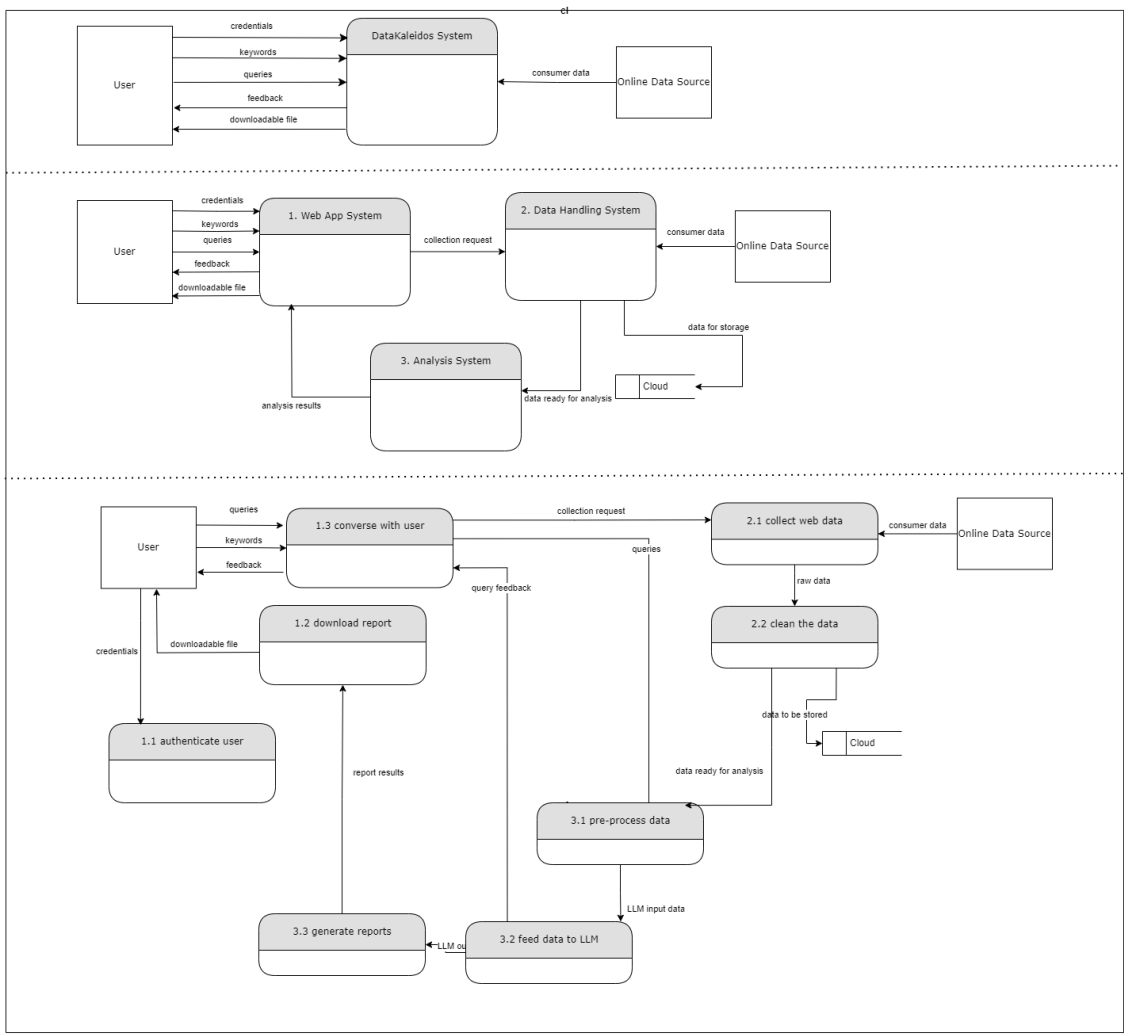


Figure 3.2: Data Flow Diagram

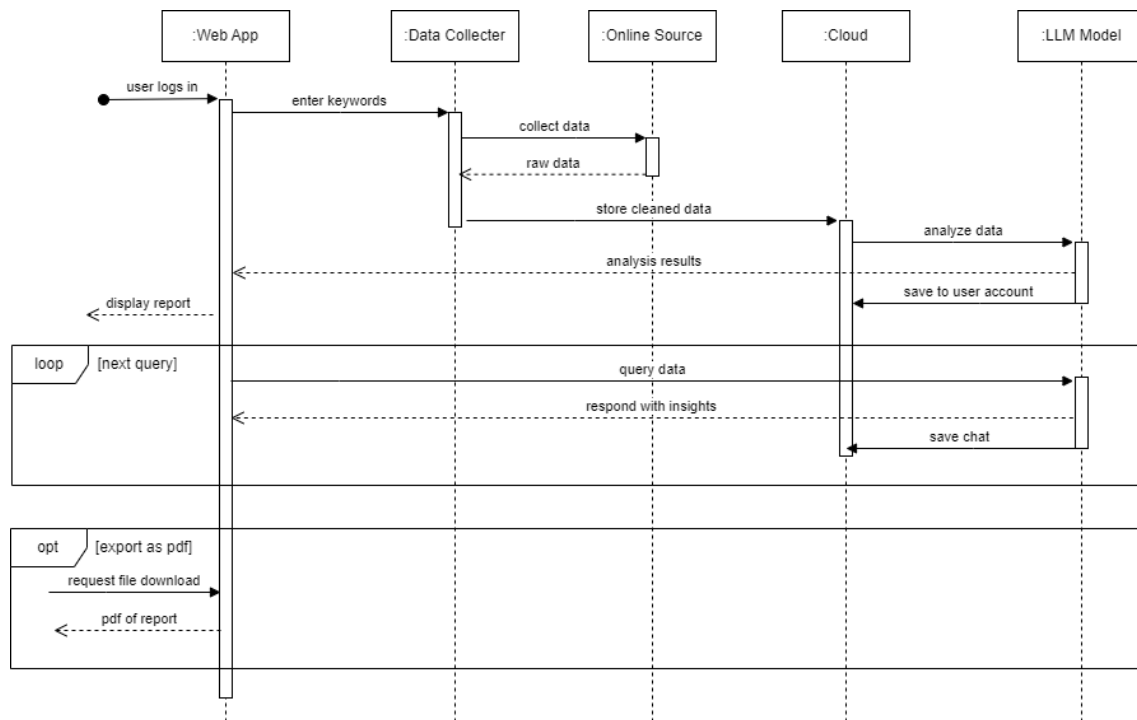


Figure 3.3: System Level Sequence Diagram

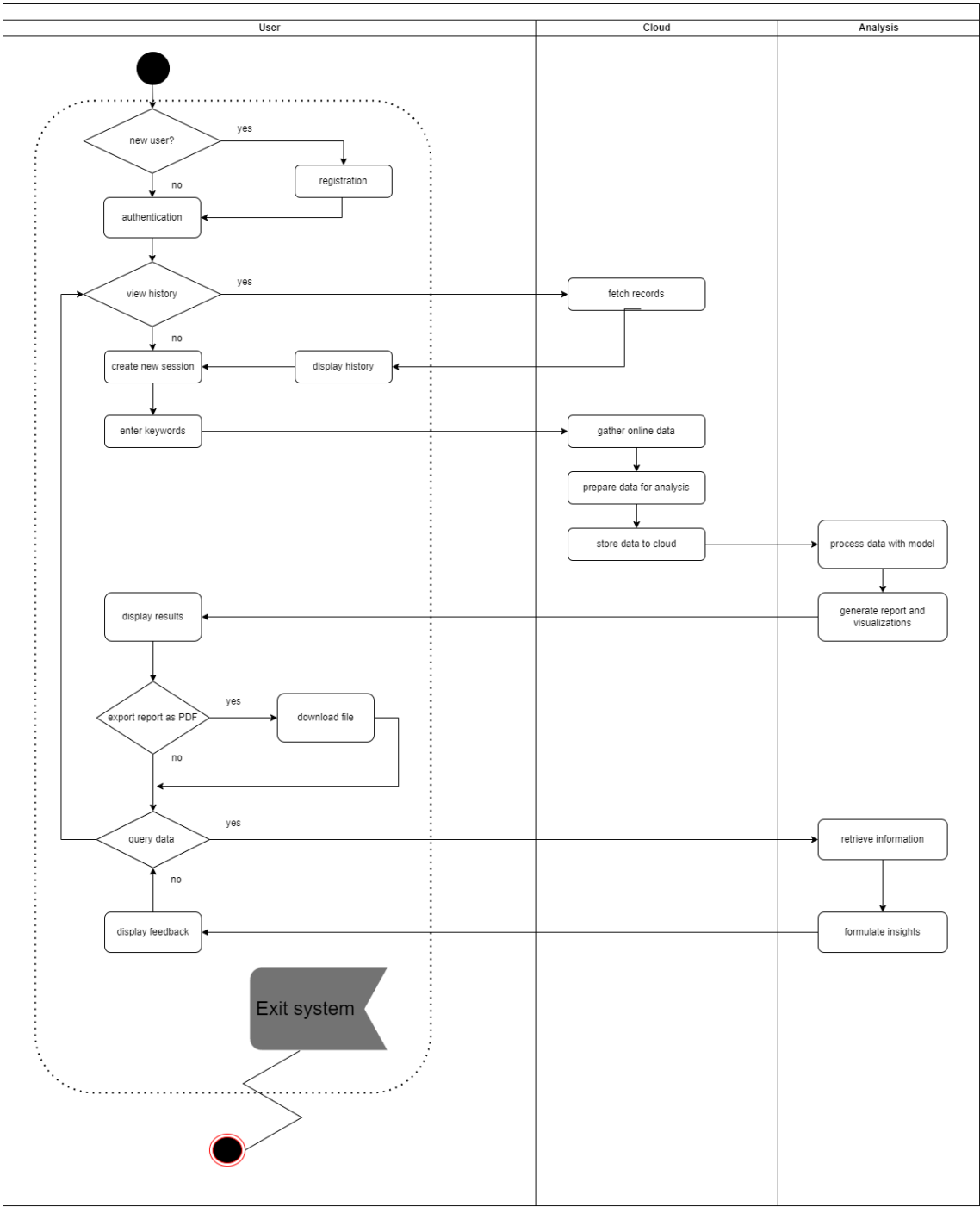


Figure 3.4: Activity Diagram

3.3 Data Design

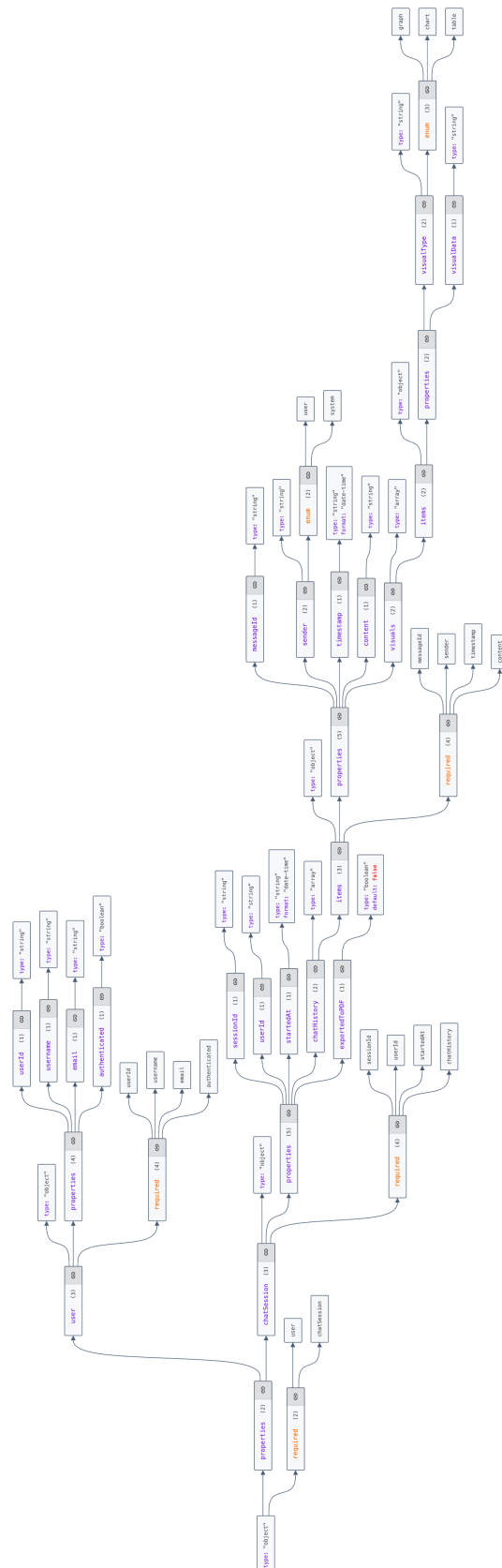


Figure 3.5: JSON Schema Diagram

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- [1] Amjad Iqbal, Rashid Amin, Javed Iqbal, Roobaea Alroobaea, Ahmed Binmahfoudh, and Mudassar Hussain. Sentiment analysis of consumer reviews using deep learning. *Sustainability*, 2022.
- [2] Social Standards. Social standards, 2024. Accessed: 2024-10-04.